We've been using the Browser to load pages

- File -> Open renders an html page
  - Does not load a page from a **server**
  - Impacts things like **absolute paths**
- You'll be writing HTML for pages on a server
  - We should use a server to validate our HTML

npx will load and run NodeJS modules

- You installed NodeJS for assignment 1
    - This also installs a program called `npx`
    - Will load and run NodeJS modules
- `serve` is a program that runs a SIMPLE webserver
    - Very basic and simple
    - "dumb"

Using npx to run serve

- At command line (terminal) run:
    - `npx serve`
- First time you do this it will ask if it should install
    - Say "y"
- Later uses will use that install

The directory you are in matters!

- Remember: All commands are run "in" a folder
- Wherever you run `npx serve`
    - THAT folder will be "**document root**"
    - The `/` of absolute paths
    - Make sure to run server at correct directory
        - "correct" based on files intended for `/`
    - Assignment has `work/02-basic-layout` as `/`

Filenames

`index.html` is a common **default file**

- If you request `/` on the server
    - Ex: **http://examplecat.com/**
    - Same as **http://example.cat/index.html**
- If you don't have any path in url
    - Browser will request `/`
        - Server may treat as `/index.html`
    - Browser may hide `/`
    - Ex: **http://examplecat.com/** shows as **http://examplecat.com**

Rules about folders in Document Root

- `http://example.com/cats/`
    - Server MAY return `http://example.com/cats/index.html`
    - Not all servers use `index.html` as default file
        - Most do
- It may be a file or a folder!
    - `http://example.com/cats`
        - Might be a file named `cats`
        - Might be a folder named `cats`

`serve` has some unusual quirks

- Removes the file extension
    - If you request `/cats.html`
        - Browser will show `/cats`
    - Most servers don't do this confusing change
        - Make sure your links include extension!
            - Except for links using default file
- If no `index.html` in requested folder
    - Will show a file listing
    - Most servers don't do this
        - Considered poor security
        - We only use `serve` to develop

Stopping the server

- As with most command line programs
    - Stop program using Ctrl-C in the terminal
    - Make sure terminal window has the focus

Some assignments use a custom webserver

- For some assignments/projects I have you run a custom webserver
    - I provide it when the time comes
- You will need to install the dependencies
    - Instructions will be in those assignments
- Otherwise, similar to `npx serve`
    - Run command as instructed
    - There will be a **document root**
    - Stop running with Ctrl-C

The `package-lock.json` file

- Outside scope of this course to cover in detail
- Used to track EXACTLY what is installed
    - For most jobs this is committed to the repo
    - In this course, just add to `.gitignore` file